MA 301

Test 3

Spring - 2014

Problem	points	out of
1		10
2		10
3		6
Score		26

Name: ____________________

HR: ____________________

Signature: ____________________

Formula Sheet

Fourier Series: $$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L} \quad \text{where} \quad d_n = \frac{1}{2L}\int_{-L}^{L} f(x)\, e^{-i\pi n x/L}\, dx$$

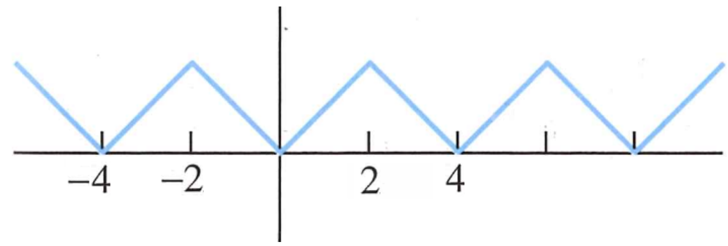
if $f(x)$ is even then $$d_n = \frac{1}{L}\int_{0}^{L} f(x)\cos\left(\frac{\pi n x}{L}\right) dx$$

if $f(x)$ is odd then $$d_n = \frac{-i}{L}\int_{0}^{L} f(x)\sin\left(\frac{\pi n x}{L}\right) dx$$

Problem 1: Consider the sawtooth function, $f(x)$, below. Determine the Fourier Series of $f(x)$ in :

(a) Exponential Form: $e^{i\pi n x/L}$

(b) Cosine & Sine Form: $\cos\left(\frac{\pi n x}{L}\right)$ & $\sin\left(\frac{\pi n x}{L}\right)$



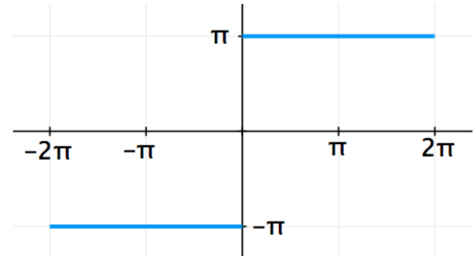
Input answers here:
(a) Exponential Form: $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/4}$ where $d_n = \begin{cases} & n = 0 \\ & n \neq 0 \end{cases}$ $a_0 =$ (b) Cosine & Sine Form: $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{\pi n x}{4}\right) + b_n \sin\left(\frac{\pi n x}{4}\right)$ where $a_n =$ $b_n =$

Problem 2: For the function depicted below, determine the Fourier Series in:

(a) Exponential Form: $e^{i\pi n x/L}$

(b) Cosine & Sine Form: $\cos\left(\frac{\pi n x}{L}\right)$ & $\sin\left(\frac{\pi n x}{L}\right)$

Write you answers in the same format as in **Problem 1**.



(assume periodic extension along the x-axis)

Input answers here:

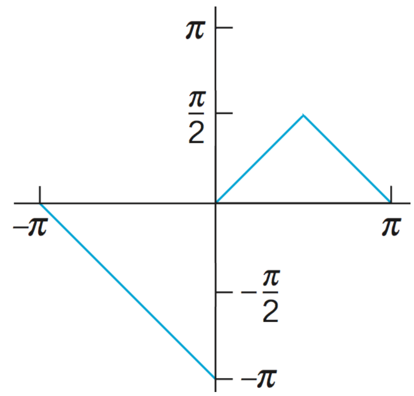
(a) Exponential Form: $f(x) =$ where $d_n = \begin{cases} & n = 0 \\ & n \neq 0 \end{cases}$

$a_0 =$

(b) Cosine & Sine Form: $f(x) =$ where $a_n =$

$b_n =$

Problem 3: ***SETUP the integral (or sum of integrals)*** needed to compute the Fourier coefficients of $f(x)$ (i.e. d_0 & d_n). Fully complete d_0 & d_n up to the point where you would begin to find the anti-derivative. Assume $f(x)$ periodic extends along the x-axis.



$d_0 =$

$d_n =$